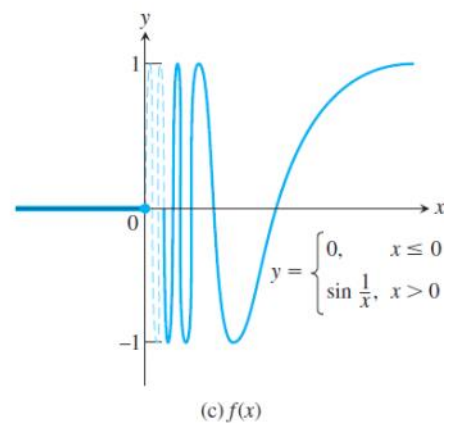
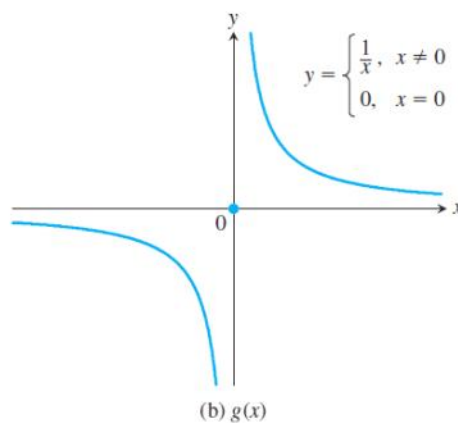
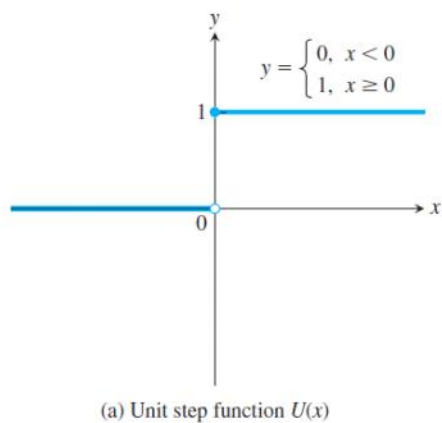
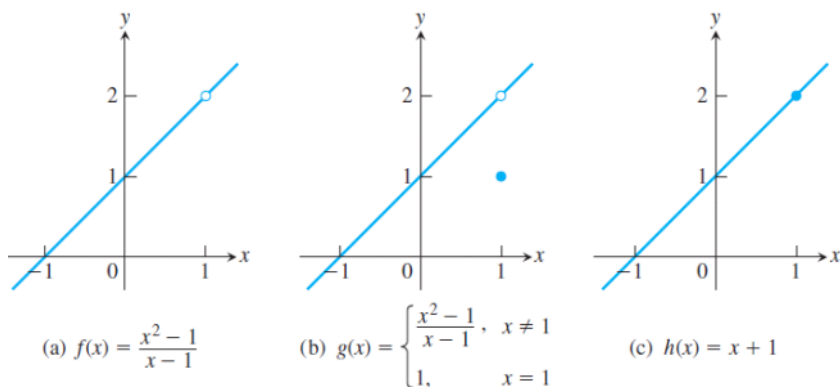


3.1 Limit(2)

Wednesday, 2 September 2026 5:18 pm



EXAMPLE 7 Evaluate

$$\lim_{x \rightarrow 1} \frac{x^2 + x - 2}{x^2 - x}.$$

EXAMPLE 8 Estimate the value of $\lim_{x \rightarrow 0} \frac{\sqrt{x^2 + 100} - 10}{x^2}.$

x	$f(x)$	
± 1	0.049876	approaches 0.05?
± 0.5	0.049969	
± 0.1	0.049999	
± 0.01	0.050000	
± 0.0005	0.050000	approaches 0?
± 0.0001	0.000000	
± 0.00001	0.000000	
± 0.000001	0.000000	

$$\begin{aligned}
\frac{\sqrt{x^2 + 100} - 10}{x^2} &= \frac{\sqrt{x^2 + 100} - 10}{x^2} \cdot \frac{\sqrt{x^2 + 100} + 10}{\sqrt{x^2 + 100} + 10} \\
&= \frac{x^2 + 100 - 100}{x^2(\sqrt{x^2 + 100} + 10)} \\
&= \frac{x^2}{x^2(\sqrt{x^2 + 100} + 10)} \\
&= \frac{1}{\sqrt{x^2 + 100} + 10}.
\end{aligned}$$

Common factor x^2

Cancel x^2 for $x \neq 0$.

Therefore,

$$\begin{aligned}
\lim_{x \rightarrow 0} \frac{\sqrt{x^2 + 100} - 10}{x^2} &= \lim_{x \rightarrow 0} \frac{1}{\sqrt{x^2 + 100} + 10} \\
&= \frac{1}{\sqrt{0^2 + 100} + 10} \\
&= \frac{1}{20} = 0.05.
\end{aligned}$$

Denominator not 0 at $x = 0$; substitute.

The Sandwich Theorem

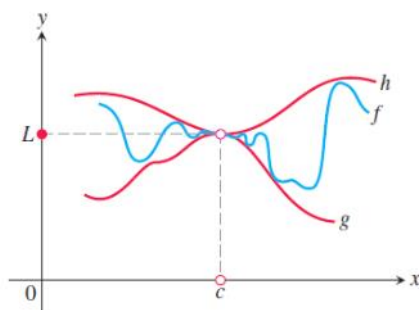
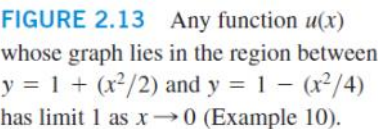


FIGURE 2.12 The graph of f is sandwiched between the graphs of g and h .

THEOREM 4—The Sandwich Theorem Suppose that $g(x) \leq f(x) \leq h(x)$ for all x in some open interval containing c , except possibly at $x = c$ itself. Suppose also that

$$\lim_{x \rightarrow c} g(x) = \lim_{x \rightarrow c} h(x) = L.$$

Then $\lim_{x \rightarrow c} f(x) = L$.


$$1 - \frac{x^2}{4} \leq u(x) \leq 1 + \frac{x^2}{2} \quad \text{for all } x \neq 0,$$
$$\lim_{x \rightarrow 0} (1 - (x^2/4)) = 1 \quad \text{and} \quad \lim_{x \rightarrow 0} (1 + (x^2/2)) = 1,$$

(a) $\lim_{\theta \rightarrow 0} \sin \theta = 0$ (b) $\lim_{\theta \rightarrow 0} \cos \theta = 1$
 (c) For any function f , $\lim_{x \rightarrow c} |f(x)| = 0$ implies $\lim_{x \rightarrow c} f(x) = 0$.

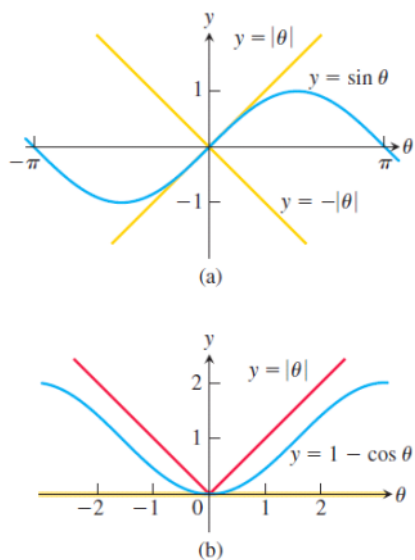


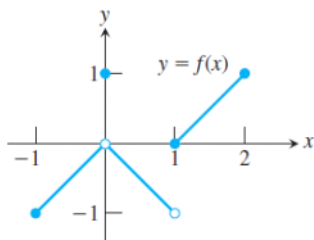
FIGURE 2.14 The Sandwich Theorem confirms the limits in Example 11.

THEOREM 5 If $f(x) \leq g(x)$ for all x in some open interval containing c , except possibly at $x = c$ itself, and the limits of f and g both exist as x approaches c , then

$$\lim_{x \rightarrow c} f(x) \leq \lim_{x \rightarrow c} g(x).$$

3. Which of the following statements about the function $y = f(x)$ graphed here are true, and which are false?

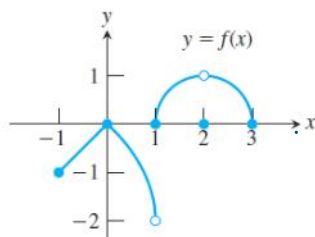
- $\lim_{x \rightarrow 0} f(x)$ exists.
- $\lim_{x \rightarrow 0} f(x) = 0$
- $\lim_{x \rightarrow 0} f(x) = 1$
- $\lim_{x \rightarrow 1} f(x) = 1$
- $\lim_{x \rightarrow 1} f(x) = 0$
- $\lim_{x \rightarrow c} f(x)$ exists at every point c in $(-1, 1)$.
- $\lim_{x \rightarrow 1} f(x)$ does not exist.



4. Which of the following statements about the function $y = f(x)$ graphed here are true, and which are false?

- $\lim_{x \rightarrow 2} f(x)$ does not exist.
- $\lim_{x \rightarrow 2} f(x) = 2$
- $\lim_{x \rightarrow 1} f(x)$ does not exist.

- d. $\lim_{x \rightarrow c} f(x)$ exists at every point c in $(-1, 1)$.
 e. $\lim_{x \rightarrow c} f(x)$ exists at every point c in $(1, 3)$.



Using the Sandwich Theorem

63. If $\sqrt{5 - 2x^2} \leq f(x) \leq \sqrt{5 - x^2}$ for $-1 \leq x \leq 1$, find $\lim_{x \rightarrow 0} f(x)$.

64. If $2 - x^2 \leq g(x) \leq 2 \cos x$ for all x , find $\lim_{x \rightarrow 0} g(x)$.

65. a. It can be shown that the inequalities

$$1 - \frac{x^2}{6} < \frac{x \sin x}{2 - 2 \cos x} < 1$$

hold for all values of x close to zero. What, if anything, does this tell you about

$$\lim_{x \rightarrow 0} \frac{x \sin x}{2 - 2 \cos x}?$$

68. Let $g(x) = (x^2 - 2)/(x - \sqrt{2})$.

- Make a table of the values of g at the points $x = 1.4, 1.41, 1.414$, and so on through successive decimal approximations of $\sqrt{2}$. Estimate $\lim_{x \rightarrow \sqrt{2}} g(x)$.
- Support your conclusion in part (a) by graphing g near $c = \sqrt{2}$ and using Zoom and Trace to estimate y -values on the graph as $x \rightarrow \sqrt{2}$.
- Find $\lim_{x \rightarrow \sqrt{2}} g(x)$ algebraically.

72. Let $F(x) = (x^2 + 3x + 2)/(2 - |x|)$.

- Make tables of values of F at values of x that approach $c = -2$ from above and below. Then estimate $\lim_{x \rightarrow -2} F(x)$.
- Support your conclusion in part (a) by graphing F near $c = -2$ and using Zoom and Trace to estimate y -values on the graph as $x \rightarrow -2$.
- Find $\lim_{x \rightarrow -2} F(x)$ algebraically.

74. Let $G(t) = (1 - \cos t)/t^2$.

- Make tables of values of G at values of t that approach $t_0 = 0$ from above and below. Then estimate $\lim_{t \rightarrow 0} G(t)$.
- Support your conclusion in part (a) by graphing G near $t_0 = 0$.

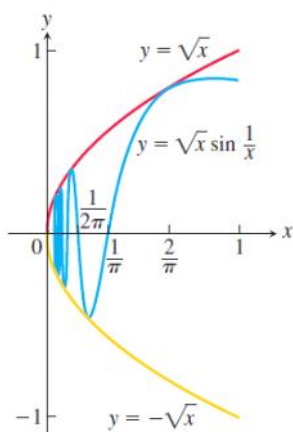
75. Let $f(x) = x^{1/(1-x)}$.

- Make tables of values of f at values of x that approach $c = 1$ from above and below. Does f appear to have a limit as $x \rightarrow 1$? If so, what is it? If not, why not?
- Support your conclusions in part (a) by graphing f near $c = 1$.

EXAMPLE 2 Show that

$$\lim_{x \rightarrow 1} (5x - 3) = 2.$$

6. Let $g(x) = \sqrt{x} \sin(1/x)$.



- Does $\lim_{x \rightarrow 0^+} g(x)$ exist? If so, what is it? If not, why not?
- Does $\lim_{x \rightarrow 0^-} g(x)$ exist? If so, what is it? If not, why not?
- Does $\lim_{x \rightarrow 0} g(x)$ exist? If so, what is it? If not, why not?

Graph the functions in Exercises 9 and 10. Then answer these questions.

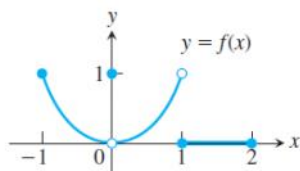
- What are the domain and range of f ?
- At what points c , if any, does $\lim_{x \rightarrow c} f(x)$ exist?
- At what points does only the left-hand limit exist?
- At what points does only the right-hand limit exist?

$$9. f(x) = \begin{cases} \sqrt{1-x^2}, & 0 \leq x < 1 \\ 1, & 1 \leq x < 2 \\ 2, & x = 2 \end{cases}$$

$$10. f(x) = \begin{cases} x, & -1 \leq x < 0, \text{ or } 0 < x \leq 1 \\ 1, & x = 0 \\ 0, & x < -1 \text{ or } x > 1 \end{cases}$$

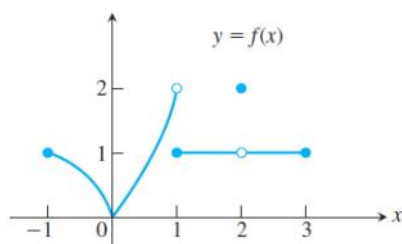
Finding Limits Graphically

1. Which of the following statements about the function $y = f(x)$ graphed here are true, and which are false?



- | | |
|---|--|
| a. $\lim_{x \rightarrow -1^+} f(x) = 1$ | b. $\lim_{x \rightarrow 0^-} f(x) = 0$ |
| c. $\lim_{x \rightarrow 0^-} f(x) = 1$ | d. $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x)$ |
| e. $\lim_{x \rightarrow 0} f(x)$ exists. | f. $\lim_{x \rightarrow 0} f(x) = 0$ |
| g. $\lim_{x \rightarrow 0} f(x) = 1$ | h. $\lim_{x \rightarrow 1} f(x) = 1$ |
| i. $\lim_{x \rightarrow 1} f(x) = 0$ | j. $\lim_{x \rightarrow 2} f(x) = 2$ |
| k. $\lim_{x \rightarrow -1^-} f(x)$ does not exist. | l. $\lim_{x \rightarrow 2^+} f(x) = 0$ |
2. Which of the following statements about the function $y = f(x)$ graphed here are true, and which are false?

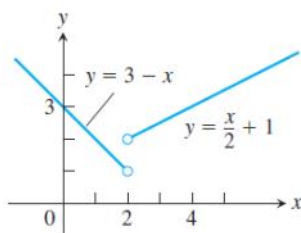
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- | | |
|---|--|
| a. $\lim_{x \rightarrow -1^+} f(x) = 1$ | b. $\lim_{x \rightarrow 2} f(x)$ does not exist. |
| c. $\lim_{x \rightarrow 2} f(x) = 2$ | d. $\lim_{x \rightarrow 1^-} f(x) = 2$ |
| e. $\lim_{x \rightarrow 1^+} f(x) = 1$ | f. $\lim_{x \rightarrow 1} f(x)$ does not exist. |
| g. $\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^-} f(x)$ | |
| h. $\lim_{x \rightarrow c} f(x)$ exists at every c in the open interval $(-1, 1)$. | |
| i. $\lim_{x \rightarrow c} f(x)$ exists at every c in the open interval $(1, 3)$. | |
| j. $\lim_{x \rightarrow -1^-} f(x) = 0$ | k. $\lim_{x \rightarrow 3^+} f(x)$ does not exist. |

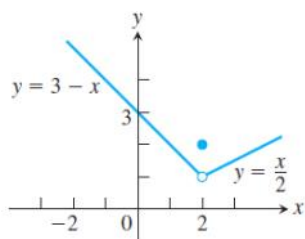
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3. Let $f(x) = \begin{cases} 3 - x, & x < 2 \\ \frac{x}{2} + 1, & x > 2. \end{cases}$



- Find $\lim_{x \rightarrow 2^+} f(x)$ and $\lim_{x \rightarrow 2^-} f(x)$.
- Does $\lim_{x \rightarrow 2} f(x)$ exist? If so, what is it? If not, why not?
- Find $\lim_{x \rightarrow 4^-} f(x)$ and $\lim_{x \rightarrow 4^+} f(x)$.
- Does $\lim_{x \rightarrow 4} f(x)$ exist? If so, what is it? If not, why not?

4. Let $f(x) = \begin{cases} 3 - x, & x < 2 \\ 2, & x = 2 \\ \frac{x}{2}, & x > 2. \end{cases}$



- Find $\lim_{x \rightarrow 2^+} f(x)$, $\lim_{x \rightarrow 2^-} f(x)$, and $f(2)$.
- Does $\lim_{x \rightarrow 2} f(x)$ exist? If so, what is it? If not, why not?
- Find $\lim_{x \rightarrow -1^-} f(x)$ and $\lim_{x \rightarrow -1^+} f(x)$.
- Does $\lim_{x \rightarrow -1} f(x)$ exist? If so, what is it? If not, why not?

7. a. Graph $f(x) = \begin{cases} x^3, & x \neq 1 \\ 0, & x = 1. \end{cases}$

- Find $\lim_{x \rightarrow 1^-} f(x)$ and $\lim_{x \rightarrow 1^+} f(x)$.
- Does $\lim_{x \rightarrow 1} f(x)$ exist? If so, what is it? If not, why not?

8. a. Graph $f(x) = \begin{cases} 1 - x^2, & x \neq 1 \\ 2, & x = 1. \end{cases}$

- Find $\lim_{x \rightarrow 1^+} f(x)$ and $\lim_{x \rightarrow 1^-} f(x)$.
- Does $\lim_{x \rightarrow 1} f(x)$ exist? If so, what is it? If not, why not?

51. **Greatest integer function** Find (a) $\lim_{x \rightarrow 400^+} \lfloor x \rfloor$ and (b) $\lim_{x \rightarrow 400^-} \lfloor x \rfloor$; then use limit definitions to verify your findings. (c) Based on your conclusions in parts (a) and (b), can you say anything about $\lim_{x \rightarrow 400} \lfloor x \rfloor$? Give reasons for your answer.

52. **One-sided limits** Let $f(x) = \begin{cases} x^2 \sin(1/x), & x < 0 \\ \sqrt{x}, & x > 0. \end{cases}$

Find (a) $\lim_{x \rightarrow 0^+} f(x)$ and (b) $\lim_{x \rightarrow 0^-} f(x)$; then use limit definitions to verify your findings. (c) Based on your conclusions in parts (a) and (b), can you say anything about $\lim_{x \rightarrow 0} f(x)$? Give reasons for your answer.

Calculating Limits

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$$17. \lim_{x \rightarrow -1/2} 4x(3x + 4)^2$$

$$19. \lim_{y \rightarrow -3} (5 - y)^{4/3}$$

$$21. \lim_{h \rightarrow 0} \frac{3}{\sqrt{3h + 1} + 1}$$

$$32. \lim_{x \rightarrow 0} \frac{\frac{1}{x-1} + \frac{1}{x+1}}{x}$$

$$34. \lim_{v \rightarrow 2} \frac{v^3 - 8}{v^4 - 16}$$

$$36. \lim_{x \rightarrow 4} \frac{4x - x^2}{2 - \sqrt{x}}$$

$$41. \lim_{x \rightarrow -3} \frac{2 - \sqrt{x^2 - 5}}{x + 3}$$

$$18. \lim_{y \rightarrow 2} \frac{y + 2}{y^2 + 5y + 6}$$

$$20. \lim_{z \rightarrow 4} \sqrt{z^2 - 10}$$

$$22. \lim_{h \rightarrow 0} \frac{\sqrt{5h + 4} - 2}{h}$$

$$42. \lim_{x \rightarrow 4} \frac{4 - x}{5 - \sqrt{x^2 + 9}}$$

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